

TUAN KIEU ANH

*Senior Unity Engineer —
Performance, Rendering &
Game Systems*

+84 98 930 88 67 tuanka1904@gmail.com linkedin.com/in/tuanka19 tuanka1904.github.io

SUMMARY

Senior Unity engineer, 8+ years building real-time software that holds up under load. Depth in GPU-driven performance — compute shaders, indirect rendering, CPU/GPU/memory profiling — and game systems engineering: inventory/crafting/loot, state machines, data-driven architecture. Ships end-to-end, from technical design to release. Unity Asset Store publisher.

HIGHLIGHTED PROJECTS

GPU Ecosystem Simulation — 2,000,000 Agents *Technical demo · Unity, Compute Shaders*

- Real-time predator-prey ecosystem simulating 2,000,000 agents on GPU at interactive framerate: compute shader behaviour updates, GPU spatial hashing for neighbour queries, one indirect draw call for rendering, zero GameObjects on the hot path.
- Built to map Unity's actual performance ceiling and the architecture required to work past it.

CatfeShader — URP Post-Processing Suite *In preparation · Unity, HLSL*

- 24 post-processing effects across two packs (Retro FX, Gameplay), authored against real frame budgets rather than screenshot quality. Third pack in design.
- Designed, built and documented solo. Preparing for Unity Asset Store release.

Catfe Vault — RPG Inventory / Crafting / Loot Framework *Submitted, pending release · Unity, C#*

- Modular RPG systems toolkit: grid inventory, crafting, loot tables, trading, storage, and a procedural affix generation module ("Forge") — fully data-driven via ScriptableObjects with custom Editor tooling.
- Five systems demonstrated in a single integrated scene where they run together: loot an item, use it, craft with it, trade it, store it.
- Architected for reuse and extension (assembly definitions, clean runtime/editor separation), with split/merge and multi-cell item support. Designed, built and documented solo; validated through external beta with independent developer testers.

Quick Scene Switcher — Unity Editor Tool *Live on the Asset Store · Unity, Editor Scripting*

- Editor extension for fast scene navigation in multi-scene projects. Released and maintained, with a free tier alongside the paid Pro version.

WORK EXPERIENCE

Software & Game Developer — Ripple Interactive

2025 – Present

- Build gameplay systems, modular architectures, and custom Editor toolsets in Unity (C#) to maximize development efficiency and component reuse.
- Deliver interactive installation and event projects end-to-end — Magic Mirror, photobooth, a temple-run style game, and motion-controlled games including SMU Voice and SMU Running.
- Integrated VOD SDK (VideoOne / BytePlus) into production.
- Own features end-to-end: technical design, implementation, performance optimization, tooling, deployment.

Software Engineer — Dreamchasers Interactive

2020 – 2025

- Engineered gameplay systems for a midcore action roguelike and client projects — AI navigation, combat, boss state machines — in Unity (C#) and Unreal Engine (C++/Blueprints).
- Built networked multiplayer in Unreal across real-time combat and mahjong projects: state synchronisation and gameplay systems under latency.
- Delivered large-scale interactive installations and AR for public venues and live events on a cross-platform pipeline, with rendering optimizations for high-fidelity 3D on constrained hardware:
 - Vinpearl Phu Quoc — Digital Aquarium: interactive wall-projection environment plus an AR volcano interaction module.
 - Singapore Discovery Centre, HomeTeam NS, Christmas Wonderland: AR sandbox with educational mini-games, AR mirror for virtual uniform fitting, cycling-controlled interactive game.
- Integrated PlayFab cloud backend. Profiled and optimized projectile and enemy logic for stable multi-platform performance.

Mobile Developer — FPO

2017 – 2020

Cross-platform mobile applications; performance and UI/UX under hardware constraints.

TECHNICAL SKILLSET

Unity & graphics — C#, compute shaders, indirect rendering, GPU spatial hashing, URP post-processing & shader authoring, ScriptableObject data pipelines, custom Editor tooling, frame budgeting on constrained hardware

Unreal 4/5 — C++ & Blueprints: combat, AI, animation, Niagara VFX, networked multiplayer

Engineering — design patterns (state machine, pooling, command, observer, MVC/MVP), data structures & algorithms, assembly definitions, data-driven architecture

Languages & tools — C#, C++, Python, JavaScript, Swift · Git, Perforce, Rider / Visual Studio

Backend — PlayFab (auth, player data, cloud services), Node.js

EDUCATION

Ho Chi Minh City University of Education — Computer Science

2019